

Safety and Reliability Training

Module 1: General Introduction

Submodule 1.C

RAMS ENGINEERING OVERVIEW

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Objectives of this training

- ✓ To understand the need of RAMS Engineering
- ✓ Get to know the four RAMS concepts, their differences and relationships
- ✓ To obtain basic knowledge of RAMS activities in a cycle of a system
- ✓ To understand RAMS methodologies evolution and future challenges



Outline

- ☐ Why RAMS
- ☐ RAMS Concepts
 - ☐ Reliability
 - ☐ Availability
 - ☐ Maintainability
 - ☐ Safety
- ☐ RAMS along the Life Cycle
- ☐ History of RAMS Engineering



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Why RAMS?

Chernobyl

Boeing 737 MAX

Challenger

Main lessons learned:

- To identify and analyze systems and components failures is essential
- Design interactions and Back up/emergency systems importance
- Redundancy does not solve design errors
- Requirements traceability
- Lack of communication between managers, designers and operators can be fatal
- Procedures/ limitations/ operation conditions must be clearly stated to be known by every part involved in the product life cycle
- Importance of Safety culture

→ Need of:

Systems Engineering

+

RAMS Engineering

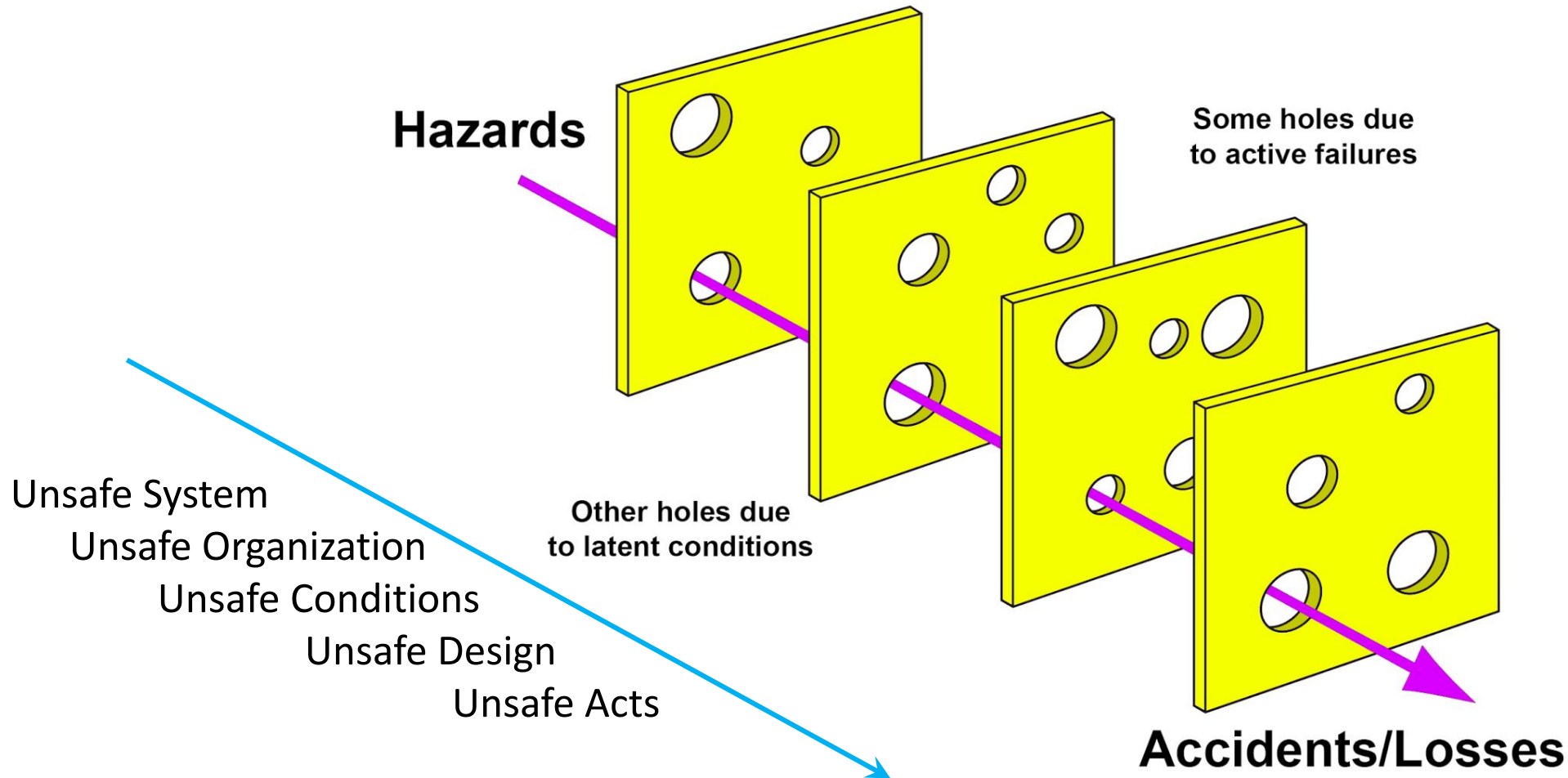
[Boeing 737 MAX accidents](#)

[Challenger O ring Final Cut – YouTube](#)

Why RAMS? Importance of RAMS Engineering

- **Regulatory Compliance:** Many industries have strict safety and reliability regulations. RAMS engineering helps organizations meet these legal requirements.
 - **Risk Management:** Identifying potential failure modes and implementing safeguards helps in proactively managing risks, leading to safer operations.
 - **Life Cycle Management:** RAMS considerations play a significant role throughout a system's life cycle, from design to decommissioning.
 - **Cost Efficiency:** By improving reliability, availability and maintainability, organizations can lower operational costs, organize logistics avoid expensive repairs or replacements.
 - **Customer Satisfaction:** Systems that are reliable and safe enhance user confidence and satisfaction, which is crucial for business success.
- Overall, RAMS engineering helps ensure that systems are effective, efficient, and safe, which is critical in today's complex operational environments.

Why RAMS? The Swiss cheese model





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RAMS Concepts: Definitions

RELIABILITY: Ability (or probability) of an item to perform a function as expected (according to established requirements/given conditions) without failure for a given time interval.

AVAILABILITY: Ability (or probability) of an item to be in a state to perform a requested function (to be operational) under given conditions at a given instant of time or over a given interval.

MAINTAINABILITY: Ability (or probability) to be retained in, or restores to, a state to perform as required, under given conditions of use and maintenance.

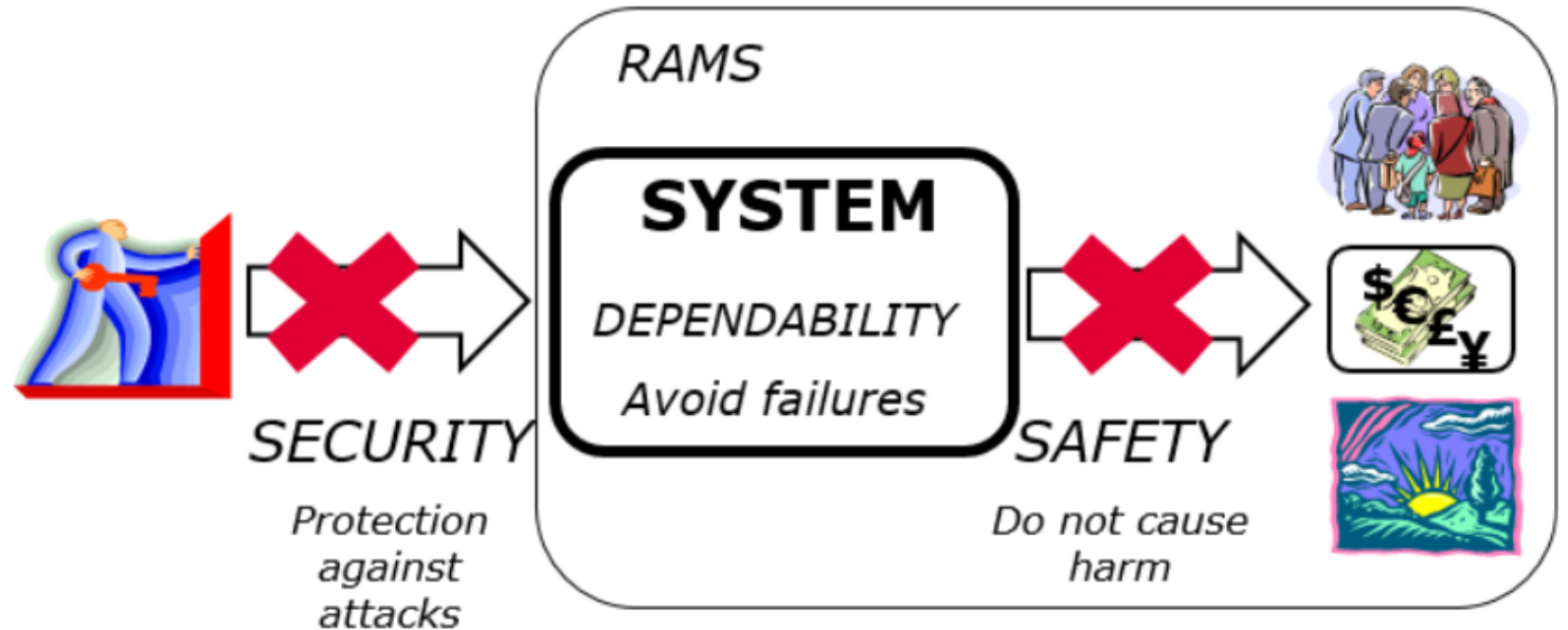
Ability to be repaired or maintained

SAFETY: Ability of a system to minimize the damage it causes and its severity.

Property whereby the risk of harm is reduced and maintained at or below an acceptable level through a continuous process of hazard identification and management/control/mitigation of risks associated with the operation.

RAMS Concepts: RAM/Safety

- **RAM** (reliability, availability and maintainability) are generally known as **dependability**.
 - Focused on avoiding failures in the system that cause its erroneous operation or a lack of service.
- **Safety**
 - Focused on preventing the system from causing harm, whether personal or environmental.
 - Safety $\neq$ Security
- ❖ Although the concepts "RAM" and Safety are different, the analyzes and methodologies (Risk Analysis, Fault Tree Analysis (FTA), FMEA / FMECA) are common to both, they are not exclusive to each area.



RAMS Concepts: Additional definitions (I)

ERROR: An omission or incorrect action by a crewmember or maintenance personnel, or a development error (e.g., mistake in requirements determination, design, or implementation).

FAILURE: An occurrence, which affects the operation of a component, part, or element such that it can no longer function as intended, (this includes both loss of function and malfunction). Note: Errors may cause failures but are not considered to be failures.

REDUNDANCY: The presence of more than one independent means for accomplishing a given function or flight operation.

PROBABILITY: A quantitative expression of the likelihood of occurrence.

RAMS Concepts: Additional definitions (II)

HAZARD: A potentially unsafe condition resulting from failures, malfunctions, external events, errors or a combination of them.

RISK: The frequency (probability) of occurrence and the associated level of hazard.

SEVERITY: The magnitude of potential consequences of a condition.

- The condition can be a failure, a combination of failures, a combination of failures with an external event,...
- Generally, the consequences considered are human death, injury, occupational illness... and damage to or loss of equipment or property. Depending on the regulation/program, it can be also covered damage to the environment, monetary loss or mission loss.

CRITICALITY: Indication of the hazard level associated with a function, hardware, software, etc., considering abnormal behavior (of this function, hardware, software, etc.) alone, in combination, or in combination with external events.

RELIABILITY

RELIABILITY: Ability (or probability) of an item to perform a function as expected (according to established requirements/given conditions) without failure for a given time interval.

- Reliability factors at item level:
 - Design/development quality.
 - Material selection.
 - Manufacturing Process.
- Reliability will always depend also on the mission profile on which the item will work.
 - Operating conditions: Temperature, humidity, loads,...
 - Usage parameters: Operating cycles, lifetime,...

AVAILABILITY

AVAILABILITY: Ability (or probability) of an item to be in a state to perform a requested function (to be operational) under given conditions at a given instant of time or over a given interval.

- Availability of an item depends on how frequently it becomes inoperable (Reliability) and how quickly it can be repaired (Maintainability).
- Availability factors:
 - Reliability
 - Maintainability
 - Maintenance Practices
 - Human Factors (training, skilled personnel availability,..)
 - Inventory Management (spare parts)
 - Monitoring and diagnostics means

MAINTAINABILITY

MAINTAINABILITY: Ability (or probability) to be retained in, or restores to, a state to perform as required, under given conditions of use and maintenance.

Ability to be repaired or maintained

- Maintainability $\neq$ Maintenance
 - Scope: Maintainability is a design consideration, while maintenance is the execution of tasks.
 - Timeframe: Maintainability is evaluated during the development phase, while maintenance occurs during the operational phase of a system.
- Two different views of Maintainability:
 - Quantitative: Measure of the speed of repairing an element or system after the occurrence of a failure (item level).
 - Qualitative: Related with accessibility, modularity, testability... (complete system level).

SAFETY (I)

SAFETY: Ability of a system to minimize the damage it causes and its severity.

Property whereby the risk of harm is reduced and maintained at or below an acceptable level through a continuous process of hazard identification and management/control/mitigation of risks associated with the operation.

- Systems shall demonstrate and acceptable level of safety against:
 - causing fatalities
 - damage to systems or facilities
 - pollution of the environment
- Safety is not a property of components, software or humans, it is a property of a **complete system** and **depends on context and environment**.



SAFETY (II)

- Safety $\neq$ Security
- Safety $\neq$ Reliability
 - Safety often means making sure X never happens
 - Reliability usually means making sure Y always happens

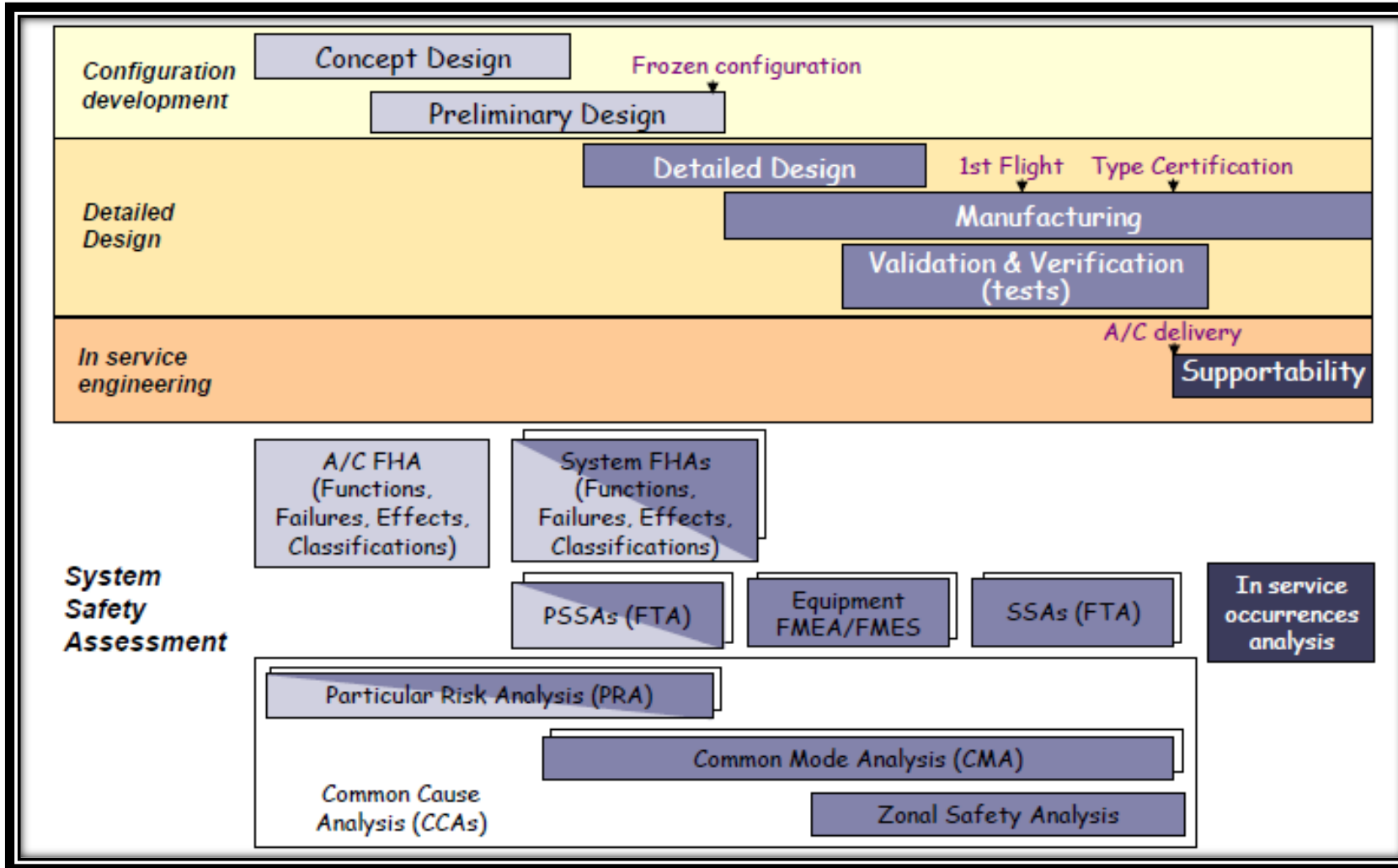
	Safe	Unsafe
Reliable	• Typical commercial flight	• Design or requirements errors. • Computer reliably executes unsafe commands. • Increasing tank burst pressure. • A nail gun without safety lockout.
Unreliable	• Aircraft engine don't start on ground. • Loss of mission functions.	• Aircraft engine fails in flight. • Failure of a critical system (flight control surface).



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- ☐ History of RAMS Engineering

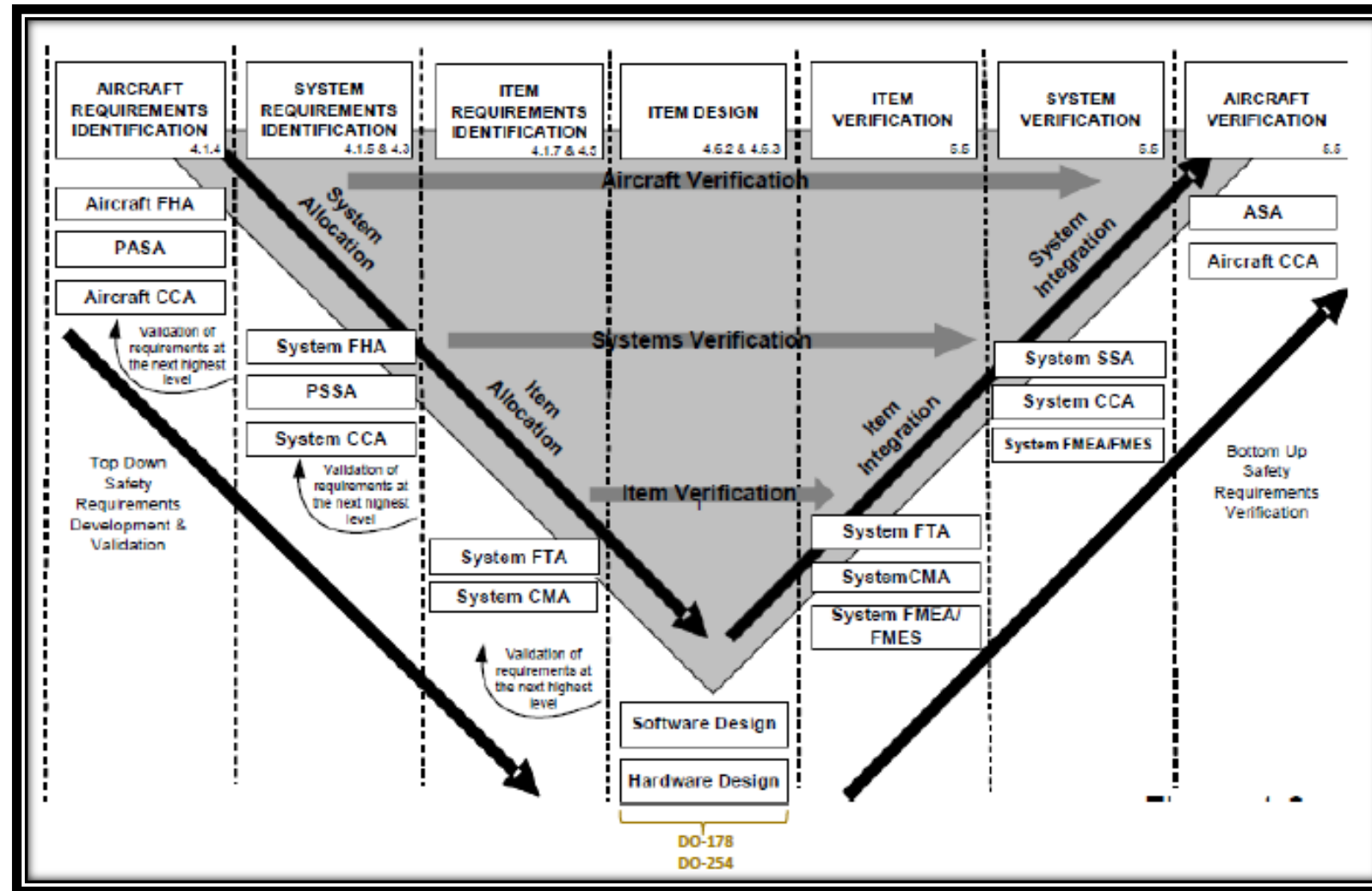
RAMS along the Life Cycle (I)



- ❖ In critical systems, RAMS analyzes are carried out in parallel with the rest of the system development, starting from the first phases and iterating with the design.
- ❖ Generally, RAMS activities are mainly concentrated in the preliminary and detailed design phases (PDR and CDR), although they are present throughout the entire life cycle.

RAMS activities in the phases of a Project

RAMS along the Life Cycle (II)



RAMS activities in the V-cycle



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- ☐ **History of RAMS Engineering**

History of RAMS Engineering (I)

1940s-1950s: Foundations of reliability engineering

- WW II: importance of improving performance and longevity of the equipment.
- Establishment of reliability testing and evaluation methods.

1960s: Formalization of Concepts

- Development of statistical methods for reliability analysis.
- Introduction of reliability metrics and failure rate calculations.

1970s: Integration of RAMS

- Increase systems complexity leads to a common engineering discipline with the four concepts.

1980s: Standardization and Methodologies

- Development of standardized methodologies, such as FMEA and FTA.
- Aerospace and defense industries adopt RAMS principles to enhance system design and safety.

1990s: Broader Application

- RAMS engineering principles expanded to other industries (automotive, telecommunications).
- Maintainability and Safety importance increase as systems became more complex.
- Enhanced Safety Regulations: Incidents and accidents led to more stringent safety regulations.

History of RAMS Engineering (II)

2000s and Beyond: Modern Developments

- Advances in technology (sw development complexity, rise of digital systems) lead to the need of the evolution of RAMS engineering.
- New methodologies and tools emerged to address challenges in highly integrated systems and cyber-physical environments.
- New considerations: Sustainability, energy efficiency and environment protection.
- Integration of security.
- Future challenges: new propulsion systems, autonomous systems, artificial intelligence...

Key Influences:

Regulatory Bodies and International Organizations:

- Promotes RAMS engineering through regulations, standards and guidelines.

Collaboration of Industry, Universities and any affected Organization:

- Research and development of RAMS methodologies covering new challenges.
- Participation and consultation in regulations and guidelines development.

Any questions?



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